Prathamesh Dalvi

Backend & Security Engineering | Seeking Internships

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SUMMARY

Third-year IT student who builds Python backends with FastAPI, Redis and SQLite, with a focus on applied cryptography, tamper-evident data and offline-first verification. I write down a threat model for each project, test it with pytest, and document its limits honestly.

EDUCATION

MCT's Rajiv Gandhi Institute of Technology (RGIT), Mumbai Expected 2028
Information Technology | CGPA: 8.32

PROJECTS

ZeroTrace - Self-Destructing Encrypted Messages Open source
Python, FastAPI, Redis, WebCrypto, Docker, Pytest | github.com/dalviprathamesh32-dev/ZeroTrace

- Encrypts messages in the browser with WebCrypto AES-256-GCM; the key stays in the URL #fragment, so the server only ever stores ciphertext.
- Stores ciphertext in Redis with a 1-hour TTL and deletes it on first read using an atomic GETDEL; a deliberate Reveal click stops link-preview bots from consuming the message.
- Validates requests with Pydantic (20,000-character limit) and tests creation, single-read deletion and 404 replay behaviour with 6 pytest tests using an in-memory fake Redis.
- Documented a threat model with explicit limitations; rate limiting, CSP and a hardened Redis config are planned for v2.

TrustLayer-CV - Provenance for Computer Vision Pipelines Smart India Hackathon (SIH26228)
Python, FastAPI, React, SQLite, Ed25519, Docker | github.com/dalviprathamesh32-dev/SIH26228-TrustLayer-CV

- Built the full prototype (backend, UI and demo): an offline-first layer that issues Ed25519-signed provenance receipts for CV inference, with nonce-based replay protection.
- Implemented an append-only, hash-chained SQLite audit ledger (SHA-256) so edits or deletions are detectable.
- Added model integrity checks (digest vs. trusted registry, behavioural fingerprint, backdoor probe), dataset poisoning signals and a tamper-simulation sandbox; covered by a 22-test pytest suite.

VAJRA - Email Threat & Forensic Intelligence Platform Smart India Hackathon (SIH26106)
Python, FastAPI | Team of four | github.com/dalviprathamesh32-dev/SIH26106-Vajra

- Contributed to the FastAPI backend of a platform that analyzes .eml/.msg emails in memory: header and hop tracing, SPF/DKIM/DMARC checks, GeoIP origin mapping and explainable risk scoring.
- Worked on a zero-database design (bounded in-memory case store with on-demand wipe) so no email payload is written to disk.

TECHNICAL SKILLS

Languages & Backend: Python 3.10+, FastAPI, Uvicorn, Pydantic V2, REST API design

Data & Storage: Redis (TTL, atomic operations), SQLite (hash-chained ledgers)

Security & Crypto: AES-256-GCM (WebCrypto), Ed25519 signatures, SHA-256 hash chaining, email header forensics, threat modeling

Infra & Testing: Docker & Compose, Pytest, httpx, Linux (Ubuntu / WSL2), Git & GitHub